

Team Details

Name	Section	SRN
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Project Title

PromptGFM-Bio — Prompt-conditioned Graph Foundation Model for Rare-Disease Gene-Phenotype Mapping

Possible Dataset and source

1. [Protein-Protein Interaction \(PPI\) graphs \(BioGRID interaction repository\)](#)
2. [STRING database](#)

Uniqueness of the topic

1. **Prompt-Conditioned Graph Learning:** Moves beyond traditional “GNN + text feature concatenation” by using natural-language prompts as active conditioning signals during graph message passing.
2. **Dynamic Semantic Injection:** Disease/phenotype descriptions are encoded as prompt embeddings and dynamically integrated into GNN layers (via cross-attention or FiLM modulation), enabling context-aware representation learning.
3. **Task-Specific Graph Reasoning:** The same biological interaction graph produces different predictions depending on the disease prompt, allowing adaptive reasoning for diverse disease queries.
4. **Foundation-Scale Biological Backbone:** Built on pretrained heterogeneous biomedical graphs (PPI networks, gene-disease links, phenotype relations), enabling transfer learning across biomedical tasks.
5. **Rare-Disease Discovery Capability:** Supports few-shot and zero-shot gene discovery, directly addressing the long-tail problem where rare diseases have extremely limited labeled data.
6. **Triple Research Contribution:** Simultaneously advances • Graph foundation model pretraining for biology, • Deep LLM-GNN structural-semantic integration, • Explainable rare-disease candidate gene ranking for interpretable biomedical discovery.

Your Learning goal

1. Design and implement a prompt-conditioning mechanism that fuses LLM-style disease semantics into GNN message passing.
2. Develop self-supervised pretraining objectives for heterogeneous biological graphs and evaluate transfer to link-prediction/ranking tasks.
3. Rigorously measure few-shot/zero-shot gains and perform ablations (no prompt, prompt as static feature, different fusion strategies).
4. Produce interpretable subgraph evidence for top candidate genes to support domain validation. Deliverables will include reproducible code, quantitative benchmarks, and at least one biologically meaningful case study.

Reference paper, if any (provide the URL)

- [1] W. Hamilton, Z. Ying, and J. Leskovec, “Inductive Representation Learning on Large Graphs,” *NeurIPS*, 2017.
- [2] W. Hu et al., “Strategies for Pre-training Graph Neural Networks,” *ICLR*, 2020.
- [3] P. Veličković et al., “Graph Attention Networks,” *ICLR*, 2018.
- [4] A. Piñero et al., “DisGeNET: A comprehensive platform integrating gene-disease associations,” *Nucleic Acids Research*, 2020.
- [5] D. Szklarczyk et al., “STRING v11: Protein-Protein Association Networks with Increased Coverage,” *Nucleic Acids Research*, 2019.